Lab Cellpose Segmentation Platform

User Guide — Neil Lin Lab, UCLA

Getting Started: Join the Lab Network

The segmentation platform runs on a lab computer and stays available at all times as a background service — nobody needs to keep a terminal window open on that computer for it to work. To reach it from your own laptop, you'll first need to join the lab's private network, called Tailscale. This only needs to be done once per device.

1. Join the lab's Tailscale network

Open this invite link and follow the prompts to sign in and join:
https://login.tailscale.com/admin/invite/yCxEjv7YirENuneirxSr11

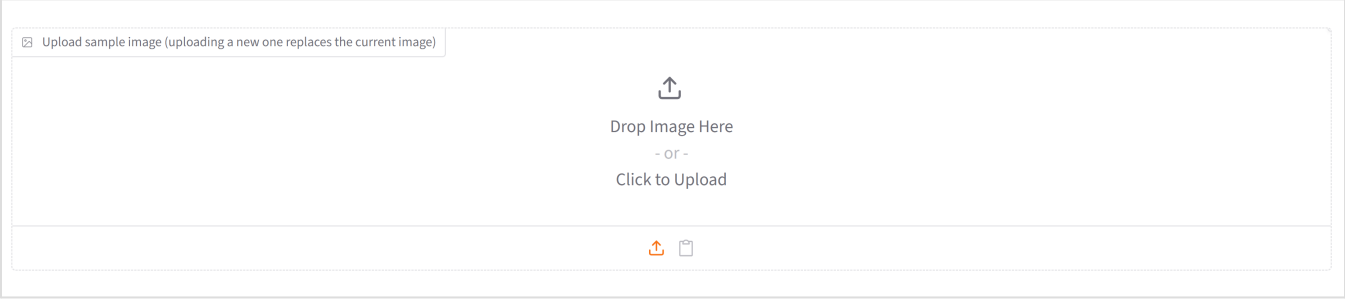
> *Tip: it's easiest to sign in with a personal email address rather than your UCLA account — the UCLA account can require extra authentication steps that get in the way.*

2. Open the platform

Once you've joined the network, open this address in your browser: http://100.111.91.76:7860. You should land on the segmentation platform's main page, shown throughout the rest of this guide.

Step 1: Set the Cell Diameter

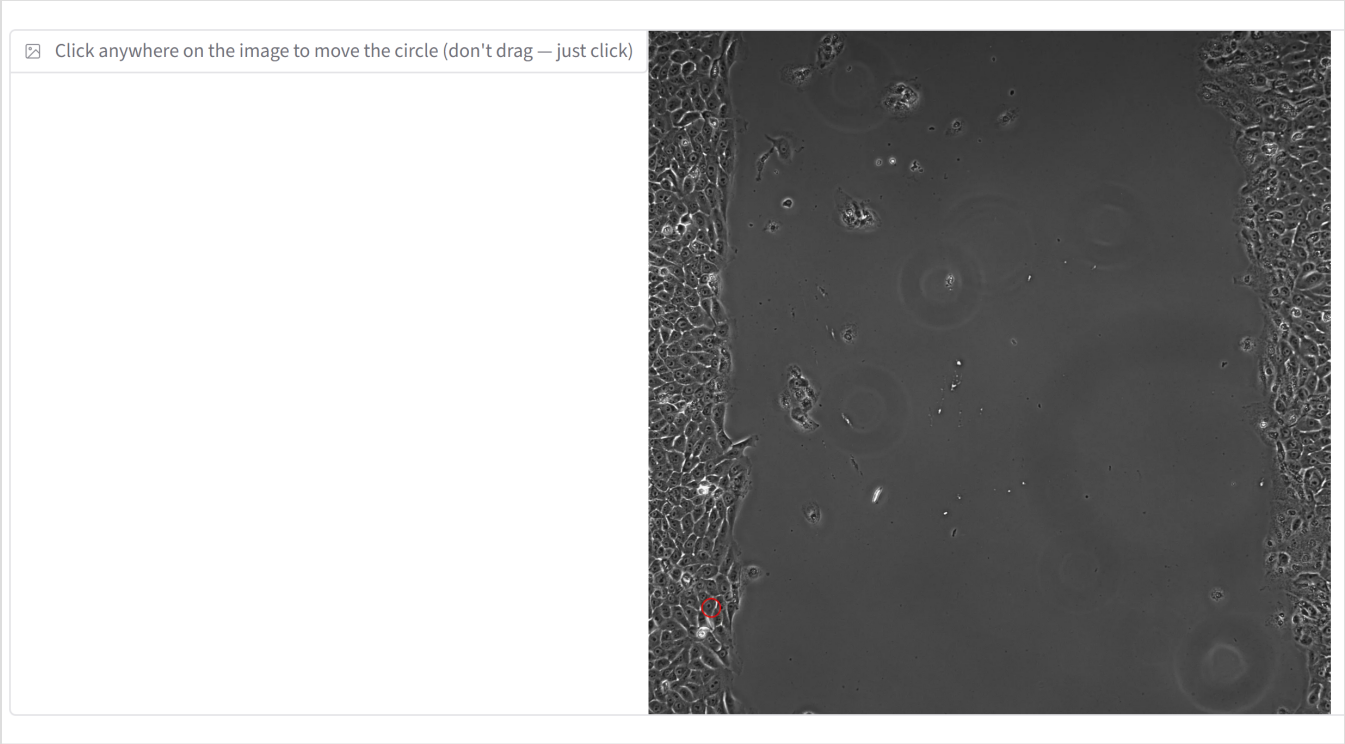
This step calibrates the typical cell size (in pixels) that Cellpose should look for, using one sample image from your batch. Upload a representative image, then click on a cell to center the red circle over it and adjust the diameter number until the circle roughly matches that cell's size.



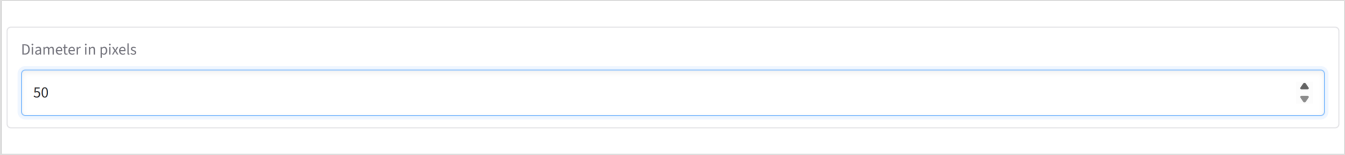
Upload a sample image here. Uploading a new one replaces the current image.



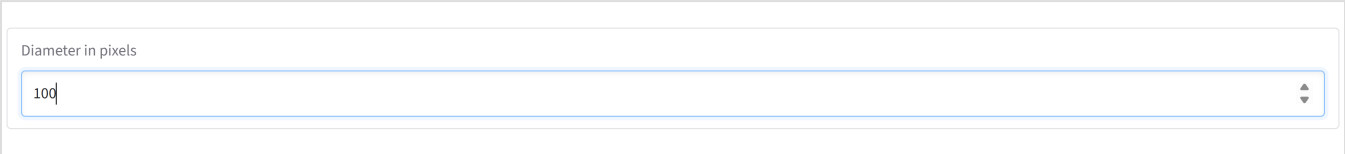
Once uploaded, the image appears with a red circle at a default position.



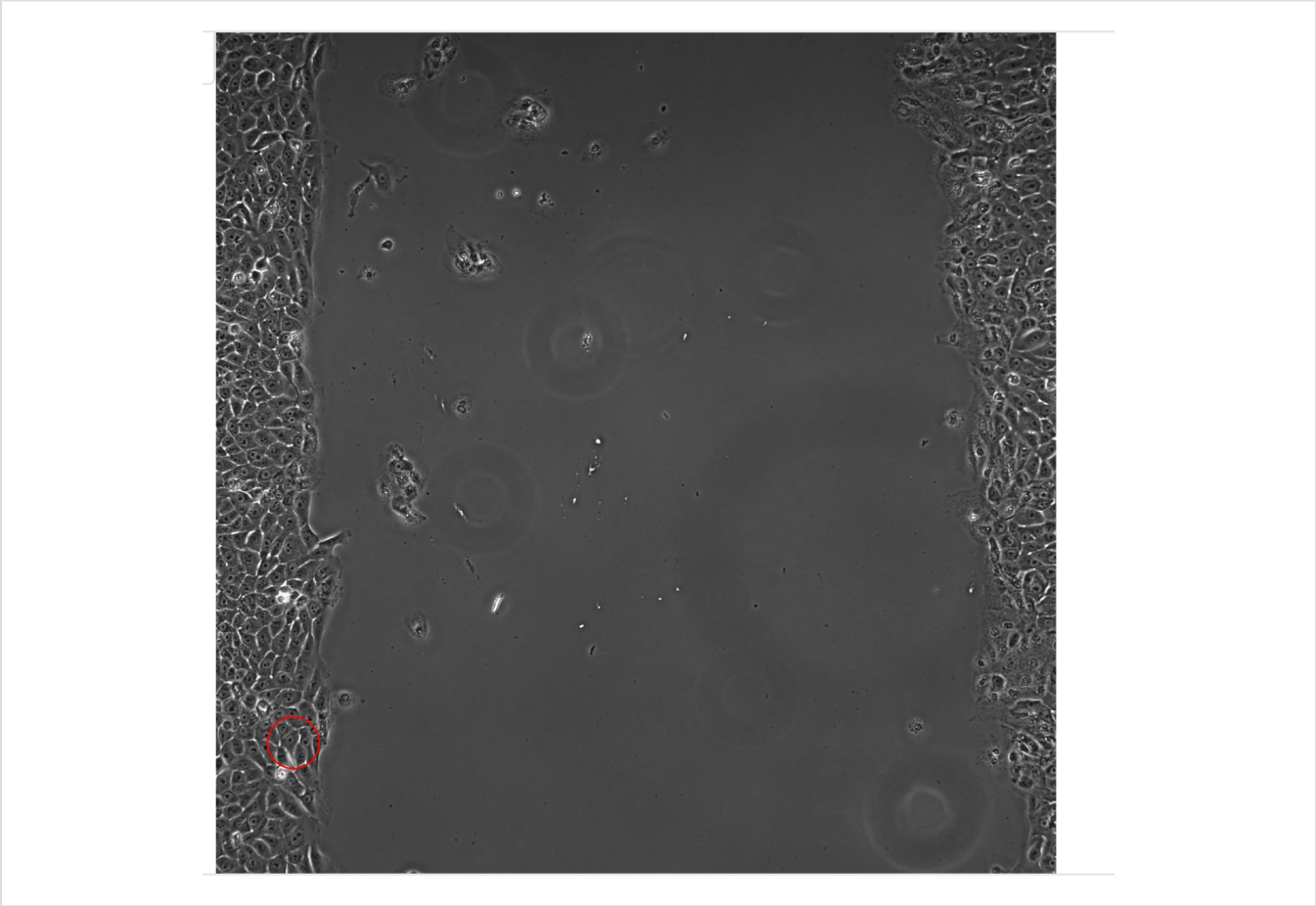
Click anywhere on the image (a single click, not a drag) to move the circle over a cell you want to size against.



The "Diameter in pixels" field controls the circle's size — shown here set to 50.



Increasing the value makes the circle larger — shown here set to 100.

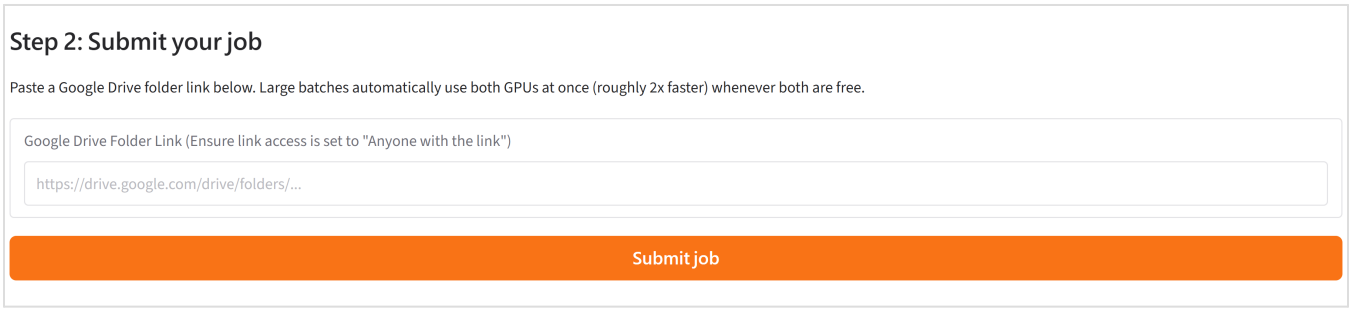


The circle scales with whatever value you enter — shown here at diameter 100. Set the diameter to whichever value best matches a typical cell in your own image, then move on to Step 2.

Step 2: Submit Your Job

Paste a Google Drive folder link containing your images, then click Submit job. Large batches automatically use both of the lab computer's GPUs at once (roughly 2x faster) whenever both are free — you don't need to do anything to enable this.

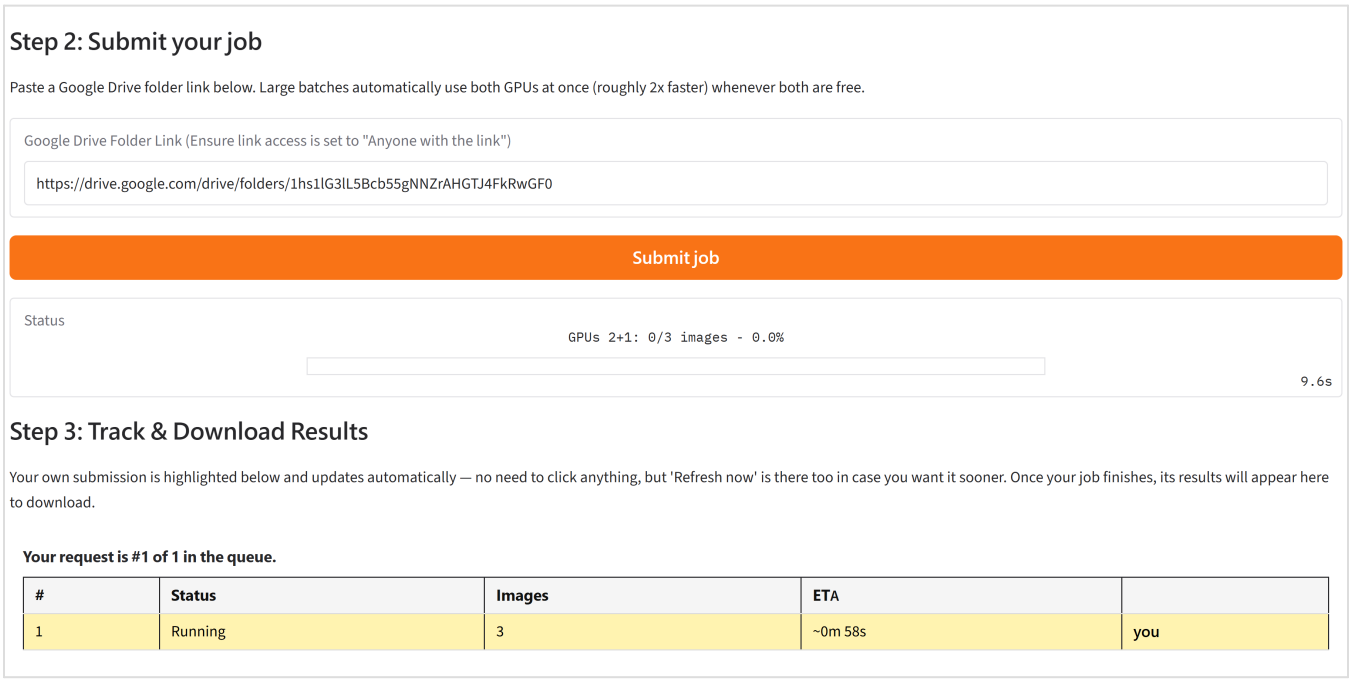
Before submitting, make sure the folder's sharing setting is set to "Anyone with the link" — otherwise the platform won't be able to download your images.



Paste your Google Drive folder link and click Submit job.

Step 3: Track & Download Results

After you submit, a status line shows live progress, and the queue table below it lists every submission currently running or waiting — your own row is always highlighted in yellow, and it updates automatically every few seconds (a Refresh now button is also available if you don't want to wait).



The status line while your job is actively running, with the queue table below it showing your row highlighted.

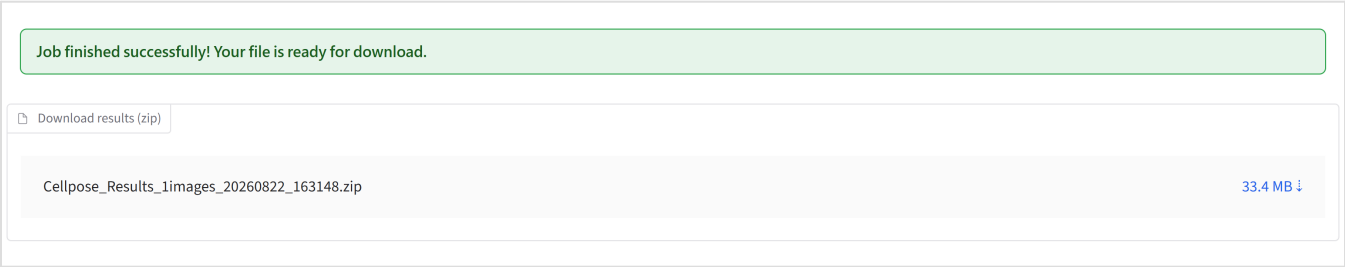


The same status bar partway through — it advances as each image finishes, here showing 2 of 3 images done (66.7%) with about 17 seconds left.

#	Status	Images	ETA	
1	Running	3	~0m 45s	
2	Queued	1	~0m 30s	
3	Queued	—	Calculating...	you

The queue table with several submissions at once — queued and running jobs are listed together, with your own request highlighted regardless of its position.

Once your job finishes, a green success banner appears above a download link for your results.



When processing completes, download your results as a single zip file.

Step 4: What's in Your Results File

Each downloaded zip is organized by well (A1, A2, ... in plate order, matching the order your images were listed in the Drive folder) and contains four things:

- **density_per_well.csv** — A summary spreadsheet with one row per well: the well name, original image filename, cell count, cell density (cells per square centimeter), and the analyzed area (in square microns).
- **debug_plots/ (one PNG per well)** — The original image with every detected cell's outline drawn on top, for a quick visual check of the segmentation quality.
- **rois/ (one ROI zip per well)** — ImageJ/Fiji-compatible ROI files for every detected cell, for anyone who wants to load the exact cell outlines into ImageJ.
- **cellpose_native/ (one set per well)** — Cellpose's own standard output files — the mask image, an outlines text file, and a seg.npy file — for anyone who wants to reopen or reprocess the results directly in Cellpose.